

Lab – Converting VDI Files for Import into Autopsy

Overview

Forensic acquisition and analysis of Virtual Desktop Infrastructure (VDI) files involves extracting data from virtual machine images and examining them for evidence relevant to a forensic investigation. VDI files are typically created and used by virtualization software such as VMware, VirtualBox, or Hyper-V.

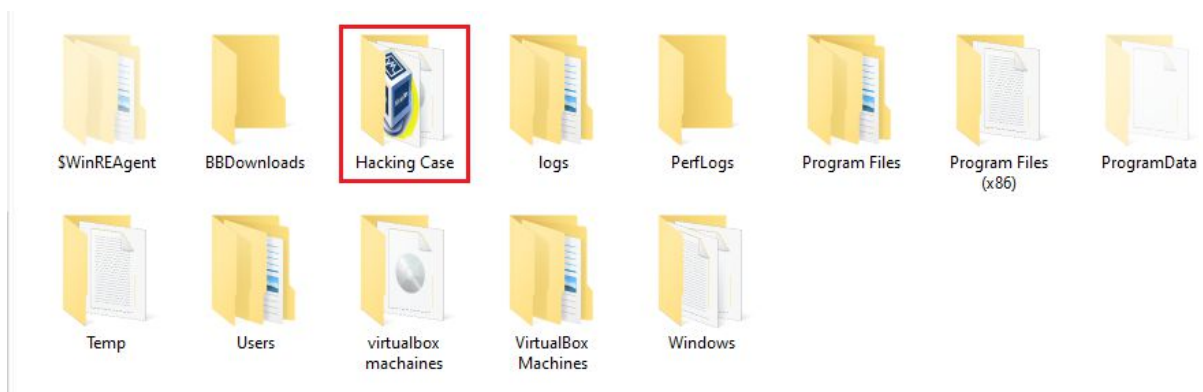
Autopsy cannot import a VDI disk. The VDI disk must first be cloned or converted as a RAW image. This is easier than it sounds using the VBoxManage disk cloning utility that comes with every installation of VirtualBox.

Lab Requirements

- One Installation of VirtualBox w/extension pack.
- One VirtualBox VDI image.
- One installation of Autopsy
- Enough disk space to save the cloned image.

Begin the lab!

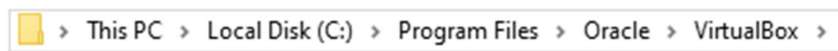
The shortest distance between any two points is a straight line. The same can be said when accessing data from the command line. In this example, I have saved a small VDI image to the root of my C: Drive in a folder labeled Hacking Case. This helps in keeping the length of the command syntax small and manageable.



Inside your C: Drive, open your folder labeled Program Files. Scroll down until you come to the folder labeled Oracle. Inside the folder, you will find a sub-directory for VirtualBox.

We next need to access the sub-director labeled VirtualBox from our command prompt. We can do this using the address bar provided for us in File Explorer. Click in the address bar of File Explorer, ensure the path is highlighted in blue, and type “cmd.” Press enter.

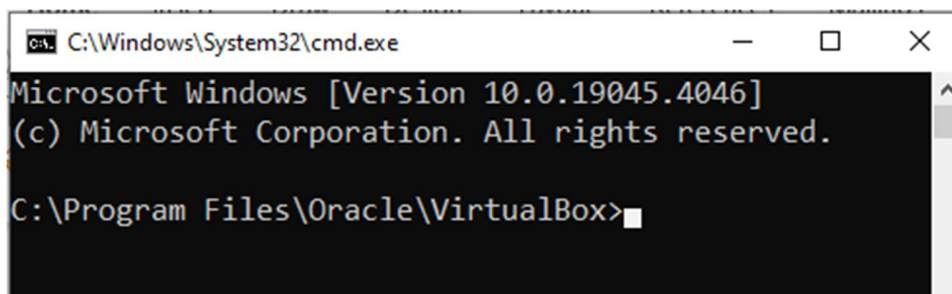
Currently, we have the following file path in our File Explorer address bar.



If you click inside the address bar, you are shown your file path, highlighted in blue. With the path highlighted, type “cmd” and press enter.



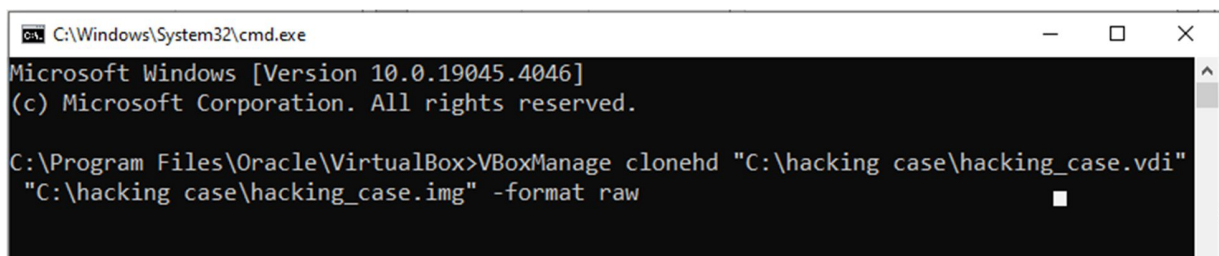
The command prompt opens at the root of the directory for VirtualBox.



Inside the VirtualBox directory is the VBoxManage cloning tool.

At the command prompt, type or copy and paste the following.

```
VBoxManage clonehd "C:\hacking case\hacking_case.vdi"  
"C:\hacking case\hacking_case.img" -format raw
```



We are telling Windows to look in the VirtualBox directory for a tool labeled VBoxManage . We then instruct VBoxManage to find the VDI image at the root of the C: Drive in a folder labeled hacking case for a VDI image labeled hacking_case.vdi and clone and save the cloned file in the same directory labeled hacking case as hacking_case.img.

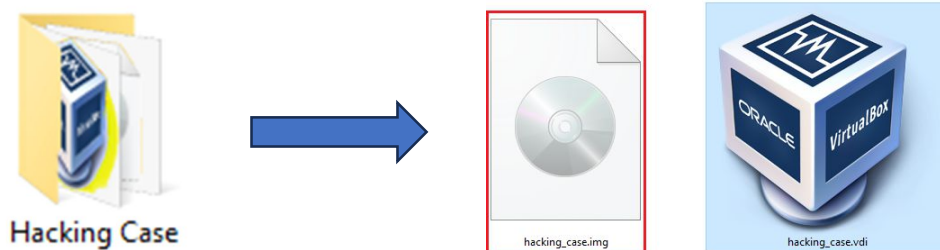
This is a tiny VDI file, so the cloning happens very quickly. This VDI image was created for instructional purposes only. A normal VDI file could be gigabytes in size, and the cloning would take some time to complete.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.

C:\Program Files\Oracle\VirtualBox>VBoxManage clonehd "C:\hacking case\hacking_case.vdi" "C:\hacking case\hacking_case.i
mg" -format raw
0%...10%...20%...30%...40%...50%...60%...70%...80%...90%...100%
Clone medium created in format 'raw'. UUID: 56d4b1ea-9346-4a74-91b6-01927bdd3a96

C:\Program Files\Oracle\VirtualBox>
```

Browse to the location where you saved the cloned image, and you should see the VDI and the RAW image in the same directory.



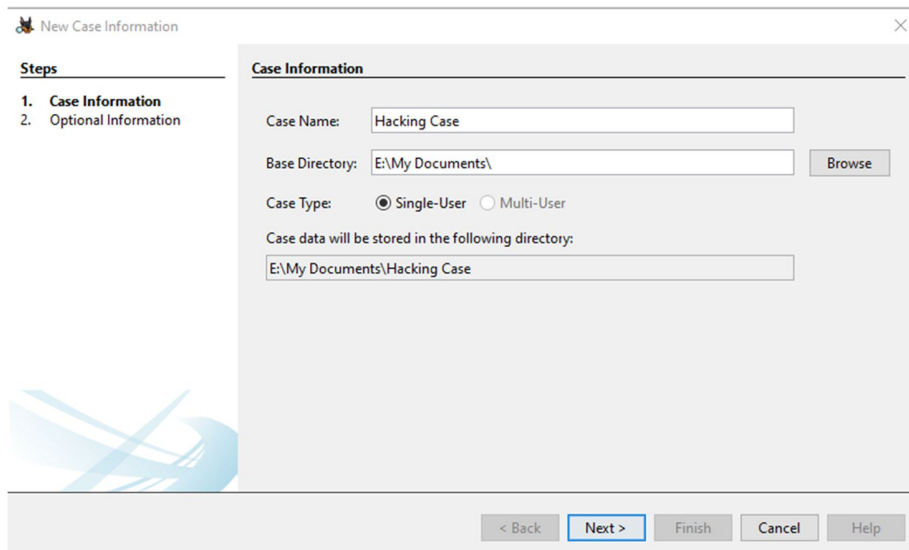
Open the cloned file using Autopsy.

Test the cloned image by creating a new case in Autopsy and importing the image to be analyzed.

Start a new case.



Give the new case a name.

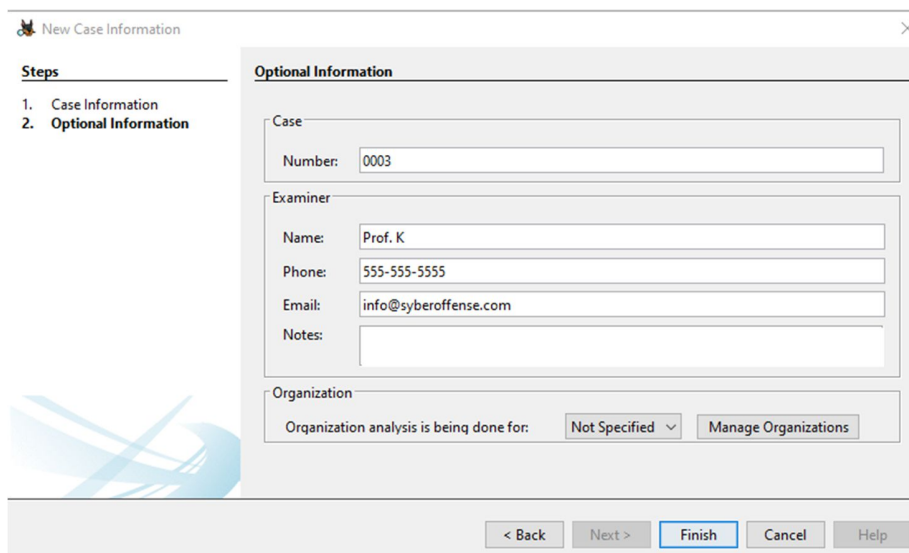


The dialog box is titled "New Case Information" and has a close button (X) in the top right corner. On the left, a "Steps" pane shows two steps: "1. Case Information" (selected) and "2. Optional Information". The main area is titled "Case Information" and contains the following fields:

- Case Name:** A text box containing "Hacking Case".
- Base Directory:** A text box containing "E:\My Documents\" with a "Browse" button to its right.
- Case Type:** Two radio buttons: "Single-User" (selected) and "Multi-User".
- Case data will be stored in the following directory:** A text box containing "E:\My Documents\Hacking Case".

At the bottom, there are five buttons: "< Back", "Next >" (highlighted in blue), "Finish", "Cancel", and "Help".

Assign the case a new case number and fill in the optional information.

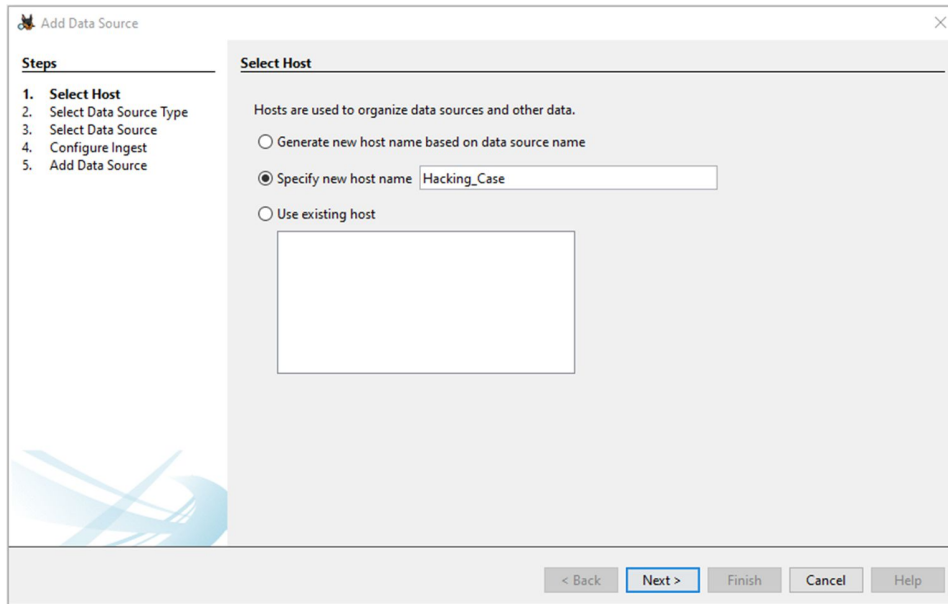


The dialog box is titled "New Case Information" and has a close button (X) in the top right corner. On the left, a "Steps" pane shows two steps: "1. Case Information" and "2. Optional Information" (selected). The main area is titled "Optional Information" and contains the following fields:

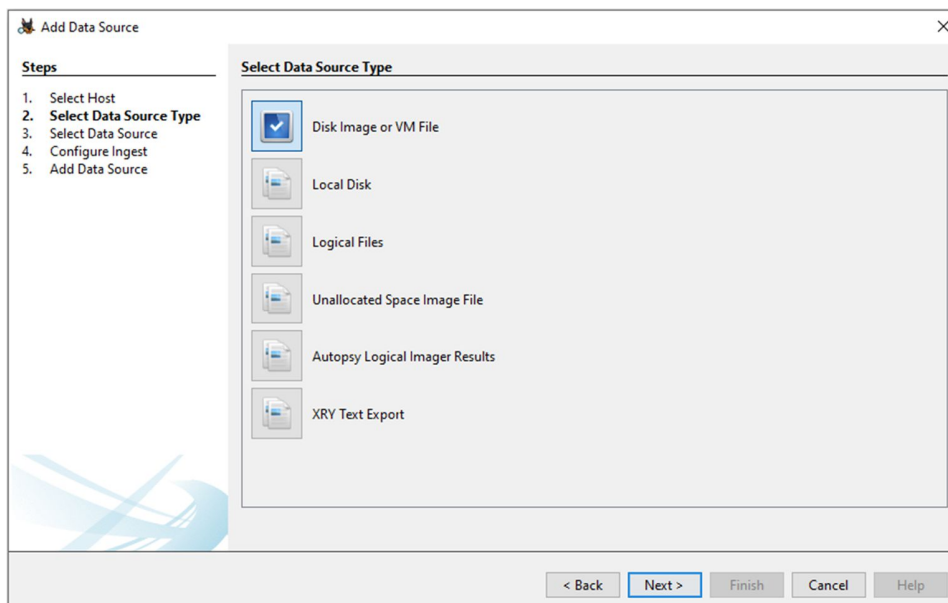
- Case:** A section containing a "Number:" text box with the value "0003".
- Examiner:** A section containing four text boxes: "Name:" (Prof. K), "Phone:" (555-555-5555), "Email:" (info@syberoffense.com), and "Notes:" (empty).
- Organization:** A section containing a text box "Organization analysis is being done for:" followed by a dropdown menu showing "Not Specified" and a "Manage Organizations" button.

At the bottom, there are five buttons: "< Back", "Next >", "Finish" (highlighted in blue), "Cancel", and "Help".

Specify a new hostname.



Select Disk Image or VM File.



Browse on over to where you save the cloned image.



Click, Next.

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
- 3. Select Data Source**
4. Configure Ingest
5. Add Data Source

Select Data Source

Path:

☐ Ignore orphan files in FAT file systems

Time zone:

Sector size:

Hash Values (optional):

MD5:

SHA-1:

SHA-256:

NOTE: These values will not be validated when the data source is added.

< Back Next > Finish Cancel Help

Accept the default for the ingest modules and click Next.

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
3. Select Data Source
- 4. Configure Ingest**
5. Add Data Source

Configure Ingest

Run ingest modules on:

☒ Recent Activity

☒ Hash Lookup

☒ File Type Identification

☒ Extension Mismatch Detector

☒ Embedded File Extractor

☒ Picture Analyzer

☒ Keyword Search

☒ Email Parser

☒ Encryption Detection

☒ Interesting Files Identifier

☒ Central Repository

☒ PhotoRec Carver

☒ Virtual Machine Extractor

Select All Deselect All History

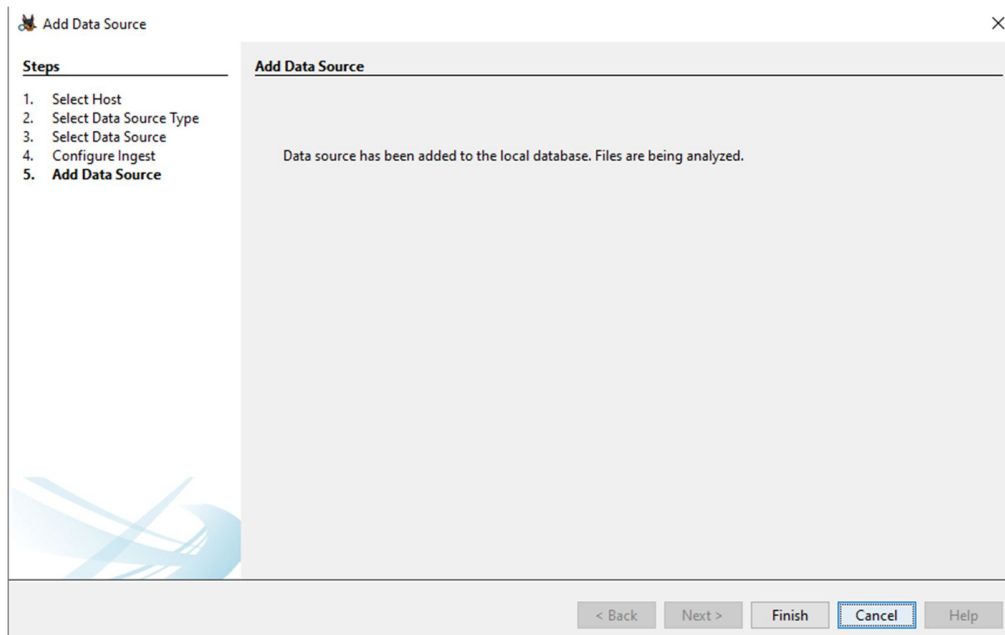
The selected module has no per-run settings.

Extracts recent user activity, such as Web browsing, re...

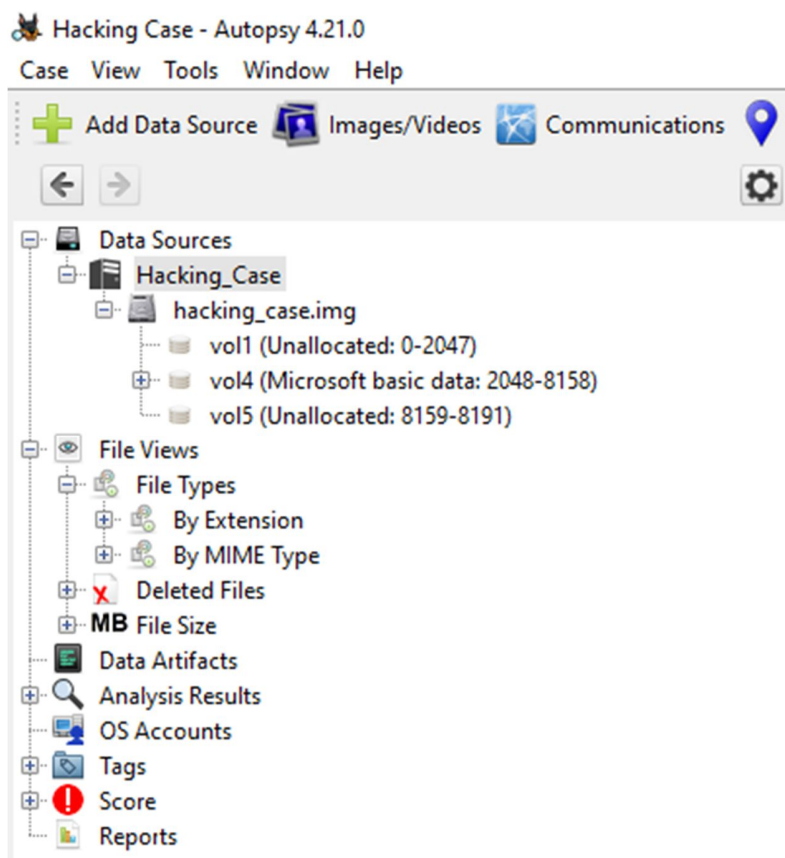
Global Settings

< Back Next > Finish Cancel Help

Click Finish.



The cloned image works!



End of the lab!